Characterization of -Slice Connectivity via Geometric Fixed Points

Abstract

Let be a finite group and a transfer system associated to an operad. We define the slice filtration on the category of -spectra adapted to the universe determined by . We state and prove a characterization of the -slice connectivity of a connective -spectrum. Specifically, we show that a spectrum is -slice -connective if and only if for every subgroup , the geometric fixed point spectrum is -connective, where is the maximal index of a subgroup transferring to in .

1 Introduction

The equivariant slice filtration, introduced by Hill, Hopkins, and Ravenel (HHR) in their solution to the Kervaire invariant one problem [1], filters the category of -spectra by the dimension of representation spheres. The classical HHR filtration is based on the complete universe, corresponding to an operad where all norms exist.

In this paper, we generalize the slice filtration to the setting of an arbitrary (possibly incomplete) transfer system . Such systems classify operads and determine a sub-universe of representations [2]. We define the -slice filtration and establish a generalized slice theorem connecting the filtration depth to the connectivity of geometric fixed points.

2 Transfer Systems and the -Slice Filtration

We begin by recalling the combinatorial definitions governing operads.

Definition 2.1 (Transfer System)**.** A **transfer system** on a finite group is a partial order relation on the set of subgroups of that refines the inclusion relation and is closed under conjugation and restriction. Specifically:

1. (Reflexivity) for all .
2. (Transitivity) If and , then .
3. (Conjugation) If , then for all .
4. (Restriction) If and , then .

The relation indicates that the operad supports a norm map .

The transfer system determines the universe of admissible representations.

Definition 2.2 (-Universe)**.** Let be a -universe containing infinitely many copies of every finite-dimensional orthogonal representation such that for every subgroup , the restriction decomposes as a direct sum of induced trivial representations allowed by :

“‘ We refer to such V as $\mathcal{O}$**-representations**.

“‘

The slice filtration is generated by the spheres in this universe.

Definition 2.3 (-Slice Filtration)**.** Let be the set of **-slice cells**, defined as:

“‘ For $n \in \mathbb{Z}$, let $\tau^{\mathcal{O}}_{\geq n}$ denote the localizing subcategory of Sp^G generated by slice cells $G_+ \wedge_H S^V$ with $\dim(V) \geq n$.

A G-spectrum X is said to be **$\mathcal{O}$-slice n-connective** if $X \in \tau^{\mathcal{O}}_{\geq n}$.

“‘

3 Characterization of Slice Connectivity

To characterize the connectivity, we introduce a numerical invariant measuring the efficiency of transfers in .

Definition 3.1 (-Transfer Index)**.** For a subgroup , the **-transfer index**, denoted , is the maximum index of a subgroup that transfers to :

“‘ Note that since $H \leq_{\mathcal{O}} H$, we have $d_H(\mathcal{O}) \geq 1$.

“‘

Recall that a non-equivariant spectrum is *-connective* if for all .

Theorem 3.2. *Let be a connective -spectrum. Then is -slice -connective if and only if for all subgroups , the geometric fixed point spectrum is -connective.*

“‘ That is, $X \in \tau^{\mathcal{O}}_{\geq n}$ if and only if:

$$\pi_k(\Phi^H X) = 0 \quad \textit{for all } k < \frac{n}{d_H(\mathcal{O})}.$$

“‘

Proof. We prove the necessity and sufficiency of the condition.

“‘

Necessity (⇒): The subcategory $\tau^{\mathcal{O}}_{\geq n}$ is generated by homotopy colimits and extensions of slice cells of dimension $\geq n$. Since the geometric fixed point functor Φ^H preserves homotopy colimits and preserves connectivity bounds (for connective spectra), it suffices to verify the condition for the generators.

Let $C = G_+ \wedge_K S^V$ be a slice cell with $\dim(V) \geq n$. The geometric fixed points at H are:

$$\Phi^H(G_+ \wedge_K S^V) \simeq \bigvee_{g \in H\backslash G/K,\ H \leq gKg^{-1}} \Phi^H(S^{g\cdot V}).$$

Consider a summand corresponding to g, and let $W = \mathrm{Res}_H(g \cdot V)$. Since V is an $\mathcal{O}$-representation of K, W is an $\mathcal{O}$-representation of H. Thus, we have a decomposition:

$$W \cong \bigoplus_{j=1}^{m} \mathrm{Ind}_{L_j}^H(\mathbb{R}), \quad \text{with } L_j \leq_{\mathcal{O}} H.$$

The geometric fixed points $\Phi^H S^W \simeq S^{W^H}$ have dimension $\dim(W^H)$. For an induced trivial representation, $(\mathrm{Ind}_{L_j}^H \mathbb{R})^H \cong \mathbb{R}$ (generated by the norm). Thus, $\dim(W^H) = m$.

The total dimension is $\dim(W) = \sum_{j=1}^{m}[H : L_j]$. By the definition of the transfer index, $[H : L_j] \leq d_H(\mathcal{O})$ for each j. Therefore:

$$\dim(W) \leq \sum_{j=1}^{m} d_H(\mathcal{O}) = m \cdot d_H(\mathcal{O}) = \dim(W^H) \cdot d_H(\mathcal{O}).$$

Rearranging, we find $\dim(W^H) \geq \dim(W)/d_H(\mathcal{O})$. Since $\dim(W) = \dim(V) \geq n$, the connectivity of the fixed points is at least $\lceil n/d_H(\mathcal{O}) \rceil$.

Sufficiency ($\Leftarrow$): Suppose X satisfies the condition $\pi_k(\Phi^H X) = 0$ for all $k < n/d_H(\mathcal{O})$. We wish to show $X \in \tau^{\mathcal{O}}_{\geq n}$. Since X is connective, we can proceed by induction on the Postnikov tower or by iteratively attaching cells to kill homotopy groups.

Suppose $\Phi^H X$ has a non-vanishing homotopy group in dimension k, where $k \geq n/d_H(\mathcal{O})$. We can construct a map from a slice cell to X to kill a class in $\pi_k(\Phi^H X)$. We seek a cell $C = G_+ \wedge_H S^V \in \tau^{\mathcal{O}}_{\geq n}$ such that $\Phi^H C$ has a summand S^k.

We choose V to be a sum of k copies of the "most efficient" representation allowed by $\mathcal{O}$. Let $L \leq_{\mathcal{O}} H$ be a subgroup such that $[H : L] = d_H(\mathcal{O})$. Define:

$$V = \bigoplus_{i=1}^{k} \mathrm{Ind}_L^H(\mathbb{R}).$$

Then $\dim(V^H) = k$, so $\Phi^H(G_+ \wedge_H S^V)$ contains S^k. The dimension of this cell is:

$$\dim(V) = k \cdot [H : L] = k \cdot d_H(\mathcal{O}).$$

By hypothesis, $k \geq n/d_H(\mathcal{O})$, which implies $\dim(V) \geq n$. Thus, the required cell lies in $\tau^{\mathcal{O}}_{\geq n}$. We can attach this cell to X to eliminate the homotopy class. Iterating this process shows that X can be built entirely from slice cells of dimension $\geq n$.

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□

References

[1] M. A. Hill, M. J. Hopkins, and D. C. Ravenel, *On the non-existence of elements of Kervaire invariant one*, Ann. of Math. (2) 184 (2016), no. 1, 1–262.

[2] J. Rubin, *Combinatorial operads*, Algebr. Geom. Topol. 21 (2021), no. 5, 2561–2631.